

TRuST-M: Evaluating User Trust and Explainability in LLM-Based Web Moderation Systems

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Abstract

As online social platforms face increasing challenges in moderating nuanced threats across web communities, trust in automated moderation systems is critical. TRuST-M examines this problem in *web and social-network environments*, where moderation occurs under time pressure and public scrutiny. Large language models (LLMs) achieve strong classification performance but remain difficult to interpret, reducing user confidence and accountability in real-world workflows. This work introduces TRuST-M, a human-centered evaluation framework that studies how explanation methods influence trust, understanding, and perceived effectiveness in LLM-based threat moderation. The framework integrates a RoBERTa model pretrained on 1M Telegram posts and fine-tuned on 15,063 labeled messages across three classes (No Threat, Judicial Threat, Non-Judicial Threat), achieving 95.8% accuracy, weighted F1 of 0.96, and Cohen's kappa of 0.94 on a held-out set. A within-subjects study (n=31) evaluated six messages of varying complexity with predictions and three explanation methods: Integrated Gradients, LIME, and attention visualization. LIME was preferred by 58% of participants for its intuitive word-level highlights, though longer response times were noted, while attention visualizations were rated least helpful due to unclear token emphasis. Statistical analysis revealed positive correlations between explanation clarity, user trust, and confidence in moderation decisions. We frame TRuST-M as an *interpretable decision-support system for human-in-the-loop moderation*, emphasizing calibrated trust and moderator comprehension rather than model replacement. The findings show that explanation clarity and response time meaningfully shape trust and decision confidence in AI-assisted moderation, advancing transparent, usable, and trustworthy moderation tools for the social web.

CCS Concepts

• **Human-centered computing** → **User studies; Empirical studies in HCI**; • **Computing methodologies** → **Natural language processing; Machine learning**; • **Security and privacy** → **Social aspects of security and privacy; Usability in security and privacy**; • **Information systems** → **Social networks; Decision support systems**.

Keywords

Explainable AI (XAI), Human–AI interaction, Trust in AI, Content moderation, Threat detection, Large language models (LLMs), User study, Social transparency, Decision support, Social networks

ACM Reference Format:

Kamalakkannan Ravi, Hemant Surale, and Jiann-Shiun Yuan. 2026. TRuST-M: Evaluating User Trust and Explainability in LLM-Based Web Moderation Systems. In *The Nineteenth ACM International Conference on Web Search and Data Mining (WSDM Companion '26)*, February 22–26, 2026, Boise, ID, USA. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3779211.3793172>

1 Introduction

The widespread deployment of large language models (LLMs) on online platforms has created both opportunities and challenges for large-scale content moderation. These systems can rapidly classify user-generated content for misinformation, hate speech, and nuanced threats [26, 39, 66]. However, their opaque decision-making makes it difficult for moderators and end users to understand why decisions are made, raising concerns about accountability, fairness, and public trust [8, 18, 48, 53]. Research shows that users are more likely to accept or challenge automated decisions when given clear explanations [19, 22], yet transparency tools remain rare in moderation workflows, especially in safety-sensitive contexts where misclassification of political content can cause real harm [24, 33, 51, 55, 70].

Beyond accuracy, effective deployment requires *usable transparency*: auditing signals must be actionable and comprehensible to decision-makers [65]. TRuST-M approaches this need by elevating *explanation clarity* as a primary design objective. In *social-web moderation settings*, our goal is not to replace human judgment but to support it—treating explainability and trust as first-class requirements in *human-in-the-loop* decision support. This contributes to broader efforts toward reliable, transparent moderation practices in fast-moving web environments.

Many safety-critical AI applications—from clinical decision support [2, 12, 16, 30, 62] to judicial threat assessment [52]—require not only accuracy but explanations that users can interpret and act upon. In content moderation, explanations help verify AI reasoning, resolve disagreements, and calibrate trust [37, 45, 53]. Yet despite extensive work in explainable AI (XAI), few techniques have been evaluated for usability or impact in real-world moderation settings [43, 56, 59].

Aligning LLM-based moderation with human expectations is further shaped by trade-offs between accuracy, usability, and interpretability. While gradient-based saliency, perturbation-based attribution, and attention visualization are common XAI methods, evidence of their effectiveness for end users remains limited



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ACM ISBN 979-8-4007-2358-2/2026/02

<https://doi.org/10.1145/3779211.3793172>

[70, 71]. Few studies assess how users interpret these explanations or whether they meaningfully improve decision outcomes in practice [3, 41].

We address this gap with TRuST-M (Threat Reasoning and User Study of Trust in Moderation), a human-centered evaluation of explainability in LLM-based threat moderation. Our interactive Total Freedom Interface (TFI) integrates three explanation techniques—gradient saliency, local perturbation via LIME, and attention visualization—within a RoBERTa classifier for judicial and non-judicial threats. Our contributions are:

- A practical *trust-explainability evaluation framework* embedded in a live moderation interface, integrating complementary attribution views and mapping them to measured trust, usability, and perceived effectiveness.
- A within-subjects user study ($n=31$) quantifying how explanation clarity, response time, and format affect trust, usability, and moderation behavior.
- Statistical evidence linking explanation clarity to trust, and trust to perceived moderation effectiveness, independent of demographic factors.
- Actionable design guidelines: (a) prioritize low-cognitive-load visualizations; (b) ensure response times support real-time decisions; (c) explore hybrid explanation formats that combine complementary strengths.

2 Related Work

Our work builds on three interconnected research areas: (1) LLM-based threat detection in NLP, (2) explainable AI (XAI) for content moderation, and (3) human-centered evaluation of trust in AI systems. We contribute a user-facing system that integrates multiple explanation techniques, empirically measures their impact on trust and decision-making, and provides actionable design guidance for practitioners. Below, we review each area and identify the research gaps addressed in this study.

2.1 LLM-Based Threat Detection (NLP Domain)

Large language models (LLMs) have significantly advanced automated online content moderation, including detection of hate speech, misinformation, and threats [26, 39, 66]. LLMs often outperform traditional classifiers [51], yet remain susceptible to overgeneralization, cultural bias, and misclassification—particularly in politically sensitive or ambiguous contexts [23, 27, 48].

Most prior work focuses on binary or coarse-grained tasks such as hate speech [17, 68], toxic language [64], or cyber threats [6]. Far fewer studies address nuanced threat categories such as politically motivated or judicial threats [28, 32, 69]. Ravi and Yuan [52] proposed a taxonomy for legal threat classification, but adoption remains limited due to the lack of domain-specific guidance for training LLMs in sensitive threat contexts.

2.2 Explainability for Content Moderation (XAI Domain)

Explainable AI (XAI) methods aim to improve transparency in model predictions. Common techniques include gradient-based saliency maps [59], perturbation-based methods such as LIME [53],

and SHAP [43]. Attention-based explanations have been applied in NLP [70], though their interpretability remains debated [57].

Despite these advances, relatively few studies apply XAI directly to real moderation workflows. Existing work exploring user trust with post-hoc explanations [4, 44] rarely considers politically sensitive or high-stakes scenarios. Similarly, Ravi and Vela [49] used attention maps for ideology classification but did not evaluate usability or satisfaction. To date, no prior work has systematically evaluated how end users interpret diverse explanations during actual threat moderation tasks.

2.3 Human-Centered Evaluation and Trust (HCI Domain)

Interpretability is both a technical and human-centered challenge, involving usability, comprehension, and trust calibration [19, 29, 41]. Ehsan et al. [22] highlight the need for social-science-informed evaluation. In safety-critical domains such as healthcare and criminal justice, studies show that explanation format influences decision-making, disagreement resolution, and appropriate reliance on AI [10, 35].

In content moderation, comparable empirical research is scarce despite similarly high stakes [54, 58, 61]. Recent work on bias and fairness emphasizes that *usable transparency* is essential for accountable decision support [9]. Our study operationalizes this perspective by treating *trust* and *explanation clarity* as measurable constructs within interactive moderation interfaces. Trust in AI is shaped by explanation complexity, message ambiguity, and user characteristics [3, 33], yet few controlled studies isolate these factors in politically sensitive threat detection. Existing findings seldom translate into actionable interface-level design recommendations [7, 20, 63].

Our focus is the *social-web moderation* setting (Telegram), where we evaluate *interpretable, human-in-the-loop decision support* for moderators.

3 Problem Statement

We evaluate how different explanation techniques influence user trust, satisfaction, and accuracy in AI-assisted threat detection and content moderation. Building on the gaps identified in *Related Work*, we formalize three research gaps:

RG 1 – Domain-Specific Threat Detection Models. We develop a custom pretrained and fine-tuned RoBERTa-based classifier for nuanced threats (judicial and non-judicial), incorporating targeted task framing, dataset design, and labeling strategies to reduce ambiguity and improve interpretability (see *Task, Data, and Model*).

RG 2 – Real-World Evaluation of XAI in Moderation. We conduct a within-subject user study using the Total Freedom Interface (TFI) to examine how explanation complexity, message ambiguity, and user characteristics affect trust, satisfaction, and classification accuracy in high-stakes moderation scenarios (see *User Study Experiment*).

RG 3 – Interface-Level Design Guidance for Trust. We analyze behavioral data and participant feedback to derive actionable design guidelines for trustworthy moderation interfaces, linking empirical findings to deployment-oriented strategies (see *Evaluation Results*).

By addressing these gaps, we assess how explanation methods shape trust, usability, and effectiveness in AI-assisted moderation, and we provide grounded design recommendations and limitations—expanded in *Evaluation Results* and *Discussion*. Our main contribution is an *evaluation and design framework* that operationalizes trust and explanation clarity for human-in-the-loop moderation on the social web.

4 Task, Data, and Model

To address the research gaps identified above, we designed a refined threat-classification task, compiled a domain-specific dataset, and developed a custom-pretrained and fine-tuned LLM classifier to serve as the foundation for the user study.

4.1 Task: Refined Threat Detection

Prior moderation research often relies on coarse categories such as normal, hate, or offensive speech [17]. While effective for broad filtering, such taxonomies lack the nuance needed to capture political rhetoric and implicit calls for violence that characterize contemporary online discourse [5]. Newer multi-level schemes add nuance but frequently blur boundaries between lawful and unlawful rhetoric [50].

To address this, we adopt a three-class taxonomy [52]:

- (1) **No Threat:** Strongly worded but non-harmful expressions (e.g., “Live free or die”).
- (2) **Judicial Threat:** Advocacy for legal or punitive action (e.g., “Lock her up”).
- (3) **Non-Judicial Threat:** Direct or implied calls for unlawful violence (e.g., “It’s time to start a civil war”). Mixed cases default to this class for severity.

This taxonomy provides a clearer separation between lawful-punitive and unlawful threats while remaining simple enough for consistent annotation and user interpretation in politically sensitive moderation contexts.

4.2 Data

To ensure ideological breadth, we collected Telegram channels from both far-left and far-right communities, though far-right content predominated due to higher posting volume. This imbalance may skew predictions and increase false positives for certain ideological groups, a limitation we acknowledge and discuss in our ethical considerations. Future iterations will explore channel-level stratified sampling, post-stratification reweighting, and robustness checks across ideological slices to assess stability under distributional shift.

The full dataset contains 2.3M replies, of which 15,076 were annotated into No Threat, Judicial Threat, or Non-Judicial Threat classes using criminology-informed guidelines. After refinement to resolve borderline cases, 15,063 labeled examples remained. This dataset was selected for its rich coverage of politically charged and grievance-driven discourse [11], providing a realistic testbed for evaluating moderation in high-risk contexts.

4.3 Modeling

The 15,063 labeled examples were divided into training (10,554), development (2,248), and test (2,261) splits using class-stratified

sampling. We used RoBERTa, a robust transformer architecture with strong performance across NLP tasks [42]. To capture platform-specific language and conversational patterns, we first pretrained RoBERTa on 1M unlabeled Telegram replies (12,260 steps; ≈ 137 hours), followed by fine-tuning on the labeled set for 100 epochs (≈ 168 hours).

The resulting model achieved 95.8% accuracy, weighted F1 of 0.96, and Cohen’s kappa of 0.94 on the held-out test set, demonstrating high agreement and strong reliability despite class imbalance and linguistic ambiguity. These properties provide a stable foundation for evaluating explanation quality and user trust in our subsequent user study.

5 User Study Experiment

The experiment evaluated how explainability-enhanced AI threat detection influences Trust (T), perceived Explainability (E), Usability (U), and Moderation Confidence (MC) when users classify politically sensitive web and social-network content. Specifically, we examined whether combining predictions, confidence scores, and explanations improves user understanding and decision confidence in high-stakes moderation scenarios.

Unlike prior work focused primarily on classification accuracy [31, 39], our study investigates human-AI interaction and user responses to different explanation types [22, 41, 47]. Although explainability has been examined in domains such as finance and healthcare, its role in online moderation—especially for ambiguous threats—is underexplored [1, 13]. We address this gap through an empirical evaluation of how three explanation methods shape trust, judgment confidence, and overall user experience.

Using a within-subjects design, participants classified six messages under three conditions: no explanation (baseline), a single explanation method, and all three methods. Each message belonged to one of the classes *No Threat*, *Judicial Threat*, or *Non-Judicial Threat*. After completing the tasks, participants filled out a System Usability Scale (SUS) survey and provided qualitative feedback.

5.1 Participants

We recruited 31 participants (15 female, 16 male), aged 18–44, through flyers and word-of-mouth. Six participants held PhDs, two held Master’s degrees, and five held Bachelor’s degrees. Thirteen participants had prior moderation experience, including two professional moderators. AI familiarity was high: 44% (N=14) were somewhat familiar, 34.3% (N=11) very familiar, and 19% (N=6) unfamiliar. Most had never used XAI tools (81.2%). All participants were active users of web and social-network platforms but unfamiliar with the explanation methods tested. Each participant received a \$10 gift card upon completion.

5.2 Explainable AI Interface

We developed the Total Freedom Interface (TFI), a custom web-based application implemented in Python using Streamlit v1.46.1 [60]. TFI integrates the RoBERTa threat classifier (Section 4.3) with three qualitatively different explanation techniques:

- **XAI Method 1 – Integrated Gradients** [36]: A gradient-based attribution method that computes path-integrated gradients to identify influential tokens.

- **XAI Method 2 – LIME (Local Interpretable Model-agnostic Explanations)** [53]: A perturbation-based technique that estimates feature importance via local surrogate models; intuitive but computationally more expensive.
- **XAI Method 3 – Final-Layer Attention Visualization** [67]: A visualization of last-layer attention weights. While attention is debated as a faithful explanation [34, 57, 70], it serves as a widely recognized baseline.

To avoid framing effects, these were shown (see Figure 1) under anonymized labels (*XAI Method 1–3*). In this study, the method–label mapping remained fixed; future replications will randomize these mappings.

Participants selected pre-loaded Telegram replies from the test set and triggered the model’s prediction. The classifier output one of three labels—*No Threat*, *Judicial Threat*, or *Non-Judicial Threat*—with distinct color-coded cues. Each explanation method used its own visualization (e.g., heatmaps, bar charts, token highlighting). The interface recorded total interaction time per message, serving as a coarse measure of cognitive effort.

TFI included usability accommodations such as dark mode, large fonts, and a responsive layout. At the end of the experiment, the system exported session logs for analysis.

5.3 Procedure

The experiment lasted about 30 minutes and followed a three-stage protocol:

- (1) **Onboarding and Demonstration (5 minutes)**: Participants received a brief overview of the study, viewed a live demonstration with a neutral example, and were warned about politically sensitive content. The three explanation methods were introduced at a high level.
- (2) **Task Execution (10–15 minutes)**: Participants classified six messages. For each message, they viewed the prediction and examined all three anonymized explanation methods. Interaction time was logged for each message.
- (3) **Survey and Feedback (10 minutes)**: Participants completed the System Usability Scale (SUS) and provided ratings of explainability, trust, model preference, and moderation confidence. Open-ended feedback captured suggestions for improvement.

All interaction logs and survey data were anonymized and stored on secure university-maintained systems. No personally identifiable information was collected. The anonymized dataset will be publicly released to support reproducibility.

5.4 Task Design

Each participant evaluated six pre-loaded Telegram replies—two per class (*No Threat*, *Judicial Threat*, *Non-Judicial Threat*). For each class, one message was designated “easy” and one “hard” based on annotator experience. Easy messages contained explicit threat markers and unanimous annotator agreement; hard messages contained implicit, sarcastic, or context-dependent language requiring guideline consultation. Message order was fixed for all participants to maintain consistent difficulty progression and reduce learning effects.

For each message, participants followed a standardized workflow in TFI: click *Start Timer*, select a message, trigger *Predict*, review the model’s classification, examine all three explanation methods, and click *End Timer*. Figure 1 illustrates the interface.

After completing all six messages, participants exported a CSV log of interaction times.

5.4.1 Measurements. We collected both objective and subjective measures to analyze how the three XAI methods influenced participant decision-making. Five measures were used:

- **Interaction Time**: Total time per message, from *Start Timer* to *End Timer*, including prediction and examination of all explanation methods.
- **Trust in AI**: Post-task rating of confidence in the model’s threat classification (1–5 scale).
- **Perceived Explainability**: Participant ratings of how well each explanation clarified the model’s reasoning.
- **Moderation Effectiveness**: Participant perceptions of how well the tool supported moderation decisions.
- **System Usability**: A 19-item usability assessment capturing ease of use, efficiency, and satisfaction.

In total: 6 messages $\times$ 3 explanation methods = 18 explanation interactions per participant, in addition to six time-logged prediction tasks and four categories of subjective ratings.

6 Evaluation Results

We now present the results of our user study evaluating the impact of three explanation techniques—Integrated Gradients, LIME, and attention visualization—on trust, usability, and classification effectiveness in AI-assisted threat moderation. All quantitative metrics and subjective evaluations are based on data from 31 participants; statistical tests and visualizations use the cleaned dataset described below.

6.1 Data Preparation and Cleaning

We first removed trials with incomplete sessions or invalid logs (e.g., missing *End Timer* clicks). For each participant, extreme interaction-time outliers (more than three standard deviations from that participant’s mean) were excluded to reduce the impact of distraction or disengagement. Across all sessions, fewer than 4% of trials were dropped due to data quality concerns. The final dataset comprised 31 participants $\times$ 6 messages $\times$ 3 explanation views = 558 explanation interactions, plus 186 timed classification tasks and 31 sets of post-task survey responses.

6.2 Analysis Overview

We used non-parametric chi-squared and Mann–Whitney tests to examine within-subject effects of explanation type and message difficulty on trust, explainability, and moderation confidence. When appropriate, post hoc pairwise comparisons (e.g., Tukey HSD) were used to identify differences between explanation techniques, and Greenhouse–Geisser or Huynh–Feldt corrections were applied when sphericity assumptions were violated.

For interaction time, we assessed normality using Shapiro–Wilk tests. When normality was violated, we applied a log transformation before running ANOVA and pairwise *t*-tests with Bonferroni

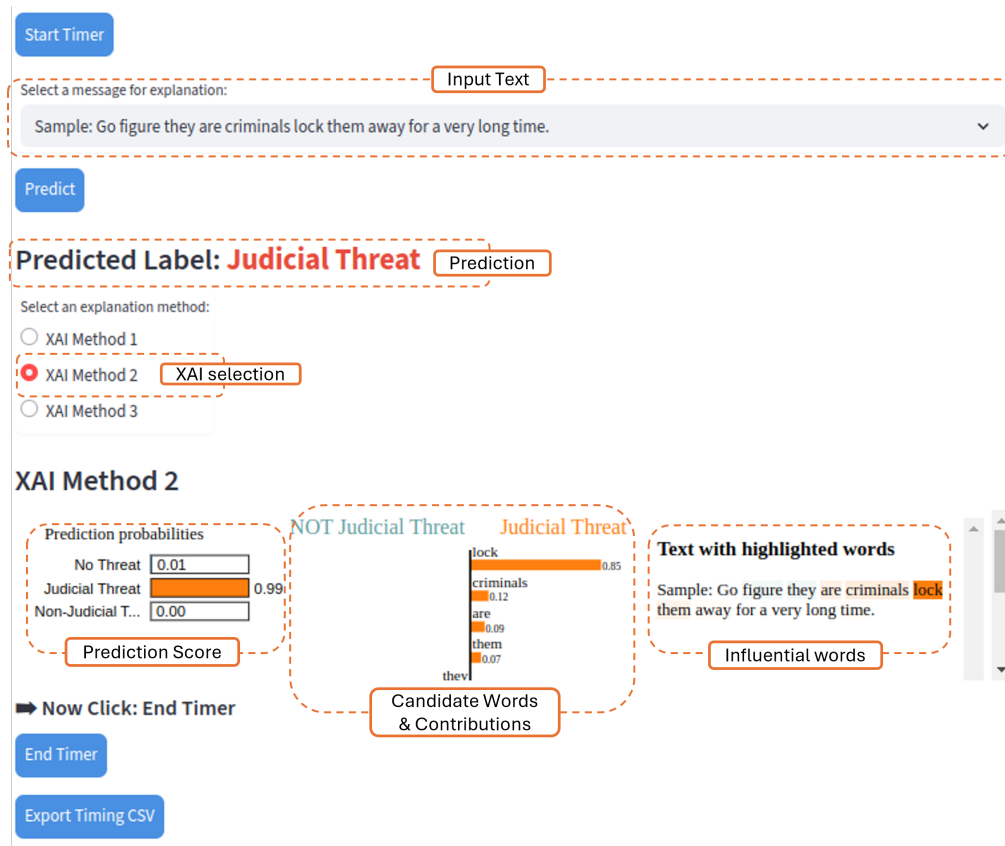


Figure 1: Annotated screenshot of the Total Freedom Interface (TFI) showing LIME-based explanation (XAI Method 2). Key components include: (Input Text), (Prediction), (XAI Selection), (Prediction Score), (Candidate Words & Contributions), and (Influential Words). Participants sequentially trigger predictions and explore explanations.

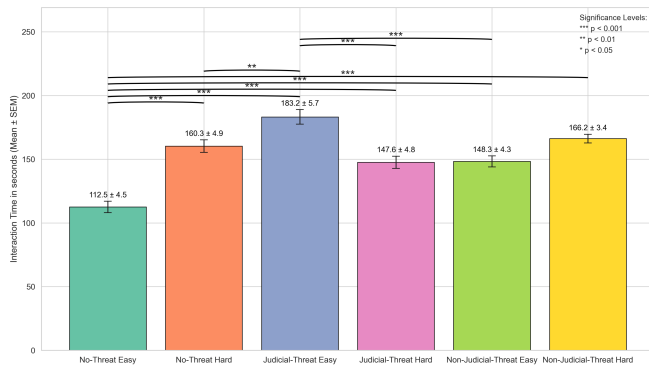


Figure 2: Mean interaction times by threat example (error bars represent standard deviation).

corrections. We used Spearman's ρ for association tests because (i) Likert-type ratings are ordinal and (ii) residuals for several measures deviated from normality; Spearman correlation is robust under these conditions.

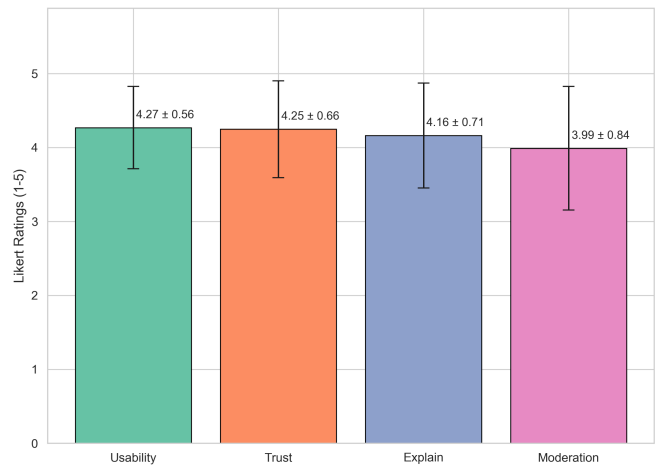


Figure 3: Mean Likert ratings aggregated by usability, trust, explainability, and confidence in moderation decisions (error bars represent standard deviation).

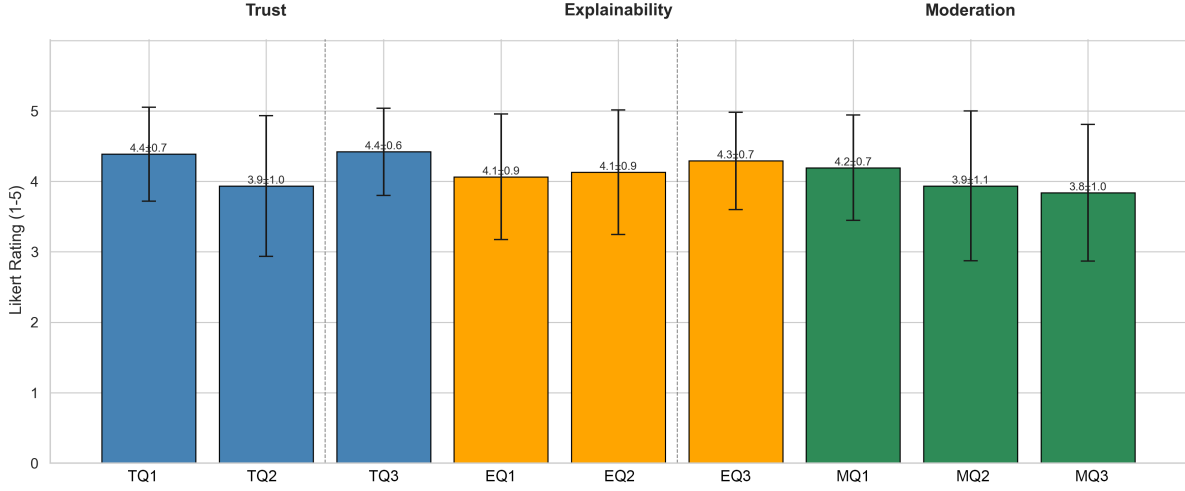


Figure 4: Aggregated user ratings for Trust (TQ1–TQ3), Explainability (EQ1–EQ3), and Moderation Confidence (MQ1–MQ3). Error bars indicate standard deviation.

6.3 Interaction Time

Average interaction time per message varied by example and difficulty (Figure 2), with statistically significant differences across pairs. As expected, “easy” tasks generally took less time than “hard” ones, except for judicial threats.

For example, an easy No-Threat task averaged 112.5 seconds, compared to 160.3 seconds for its harder counterpart—a difference of 43.8 seconds. In contrast, the “easy” judicial threat took longer than its harder counterpart, suggesting additional verification in severe cases. All differences were significant ($p < 0.05$).

The increased time on the “easy” judicial threat may reflect a confirmation–verification effect: when content appears clearly severe, participants may invest extra time to double-check the AI’s output before accepting it. This is consistent with work on confirmation bias and decision-making under emotionally charged evidence [46], as well as findings that even straightforward explanations can lead users to spend more time validating model reasoning [40].

6.4 Trust and Explainability

Subjective ratings of trust and explainability were collected on a 5-point Likert scale (5 = highest). Overall, participants reported relatively high ratings for both, with averages around 4.2. Figure 3 shows the mean trust and explainability scores aggregated across all explanation methods.

To test whether explanation clarity is associated with trust in the model, we ran a Spearman rank-order correlation. We observed a positive, statistically significant association between trust and explanation clarity ($\rho = 0.493$, $p = 0.005$), indicating a moderate, monotonic relationship (see Figure 5). In other words, participants who perceived the explanations as clearer also tended to report higher trust in the system.

6.5 Usability

Participants rated the interface as easy to understand and operate, with usability receiving the highest aggregated scores (mean = 4.27, SD = 0.56). The main complexity, according to comments, stemmed from interpreting the explanation methods themselves rather than from basic navigation or controls.

We used a 19-item standardized usability assessment to evaluate overall usability. All participants reported that the interface was easy to use and did not interfere with their interaction flow. Usability showed increasing correlations with Trust ($\rho = 0.38$, $p < 0.05$), Explainability ($\rho = 0.57$, $p < 0.01$), and Moderation Confidence ($\rho = 0.72$, $p < 0.01$), suggesting that while usability alone does not guarantee trust, it is an important enabling factor for effective decision support.

6.6 Moderation Effectiveness

Participants also rated how helpful the tool was for real-world content moderation. As shown in Figure 3, moderation effectiveness scores were slightly lower on average than trust, usability, and explainability, although these differences were not statistically significant.

Moderation effectiveness was strongly correlated with trust ($\rho = 0.62$, $p < 0.01$) and explainability ($\rho = 0.66$, $p < 0.01$), indicating that participants who trusted the model more and found explanations clearer also felt more confident using the tool for moderation decisions. Education level was associated with usability satisfaction: 83% of PhD respondents rated the system “Excellent” or “Best Imaginable,” compared to 50% of Bachelor’s degree holders who rated it “Good” or “Fair” ($\chi^2(6, N = 31) = 20.99$, $p < 0.05$). No significant effects were found for gender, age, or prior AI/ML experience.

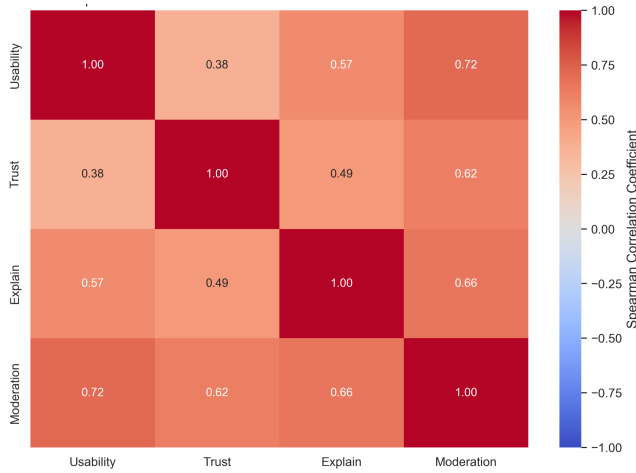


Figure 5: Spearman correlation among usability, trust, explainability, and confidence in moderation decisions (correlation coefficients shown in cells).

These patterns suggest that domain-specific familiarity and analytical comfort may influence how users experience the interface, potentially more than general AI literacy. To mitigate risks of bias or misuse, we recommend that any deployment pair this type of tool with human oversight, fairness audits, and context-specific training.

6.7 Qualitative Feedback

We conducted a thematic analysis of open-ended responses and grouped them into recurring themes.

Preference for Method 2 (LIME). Participants generally preferred simpler, color-coded visualizations over dense or text-heavy explanations. Many expressed confusion when different methods highlighted different tokens for the same message, suggesting the value of explanation alignment.

When asked which explanation method was most or least helpful, there was a strong inclination toward Method 2, which provided color-coded highlights and ranked word importance. For example, P23 noted:

“Method 2 was the most helpful, as it clearly separated the words that were important for the label given.”

Similarly, P3 commented:

“Model 2 was good because it clearly highlighted the words that were important, and it used a darker highlight to signify more importance or emphasis (which was easy to recognize), and also the explanation of ranking words from top to bottom in terms of how much they fit what the model was looking for helped understand things very clearly.”

At the same time, several participants raised concerns about response time for Method 2. As P10 put it:

“Faster XAI method 2 implementation would help [for a] real-world use case.”

Overall, 58% of participants (18/31) preferred Method 2; the remaining methods collectively accounted for the rest (~6/31).

Interest in Hybrid Explanations. A smaller subset of participants suggested that combining methods could improve interpretability. For instance, P13 remarked:

“If you combine methods 2 and 3, it would be easier for general people to interpret the outputs of the tool. Because plots are always easy and quick to evaluate and compare, and it doesn’t take a long time to read passages like what happens in method one.”

This comment about *general people* hints that explanation preferences may vary with background and familiarity, reinforcing the quantitative finding that education level influences usability perceptions.

Demographic Effects on Usability. Consistent with the quantitative analysis, qualitative responses suggested that users with more advanced education or analytical training were more comfortable with the interface and explanations. As noted earlier, education level was significantly associated with overall experience ratings ($\chi^2(6, N = 31) = 20.99, p < 0.05$). However, due to unbalanced group sizes, we could not reliably infer how prior experience with moderation tools or AI/ML systems affected trust, explainability, or moderation confidence.

Limitations of Method 3 (Attention Visualization). Participants frequently criticized Method 3 for confusing scores and unclear token importance. P22 wrote:

“XAI Method 3 was the least helpful as it did not give the more important words greater attention scores.”

Similarly, P19 and P20 raised concerns about emphasis on non-informative tokens:

“[...] XAI Method 3 was often giving higher attention to special tokens like inverted commas, which can be worked upon since they are not always the most informative tokens.” — P19

“Model 3 would constantly return high values for words that don’t seem relevant at all to the threat.” — P20

P27 suggested that adding more detail on why specific tokens were highlighted could help make this method more understandable.

Together, these findings indicate that improving the responsiveness of Method 2 and refining the interpretability of Method 3 could further enhance user experience and strengthen confidence in AI-assisted moderation.

7 Discussion

Our findings reinforce prior work in explainable AI (XAI) and human-AI interaction by showing that explanation quality directly influences user trust and perceived effectiveness in moderation tasks. In line with Ehsan et al. [22] and Liao et al. [41], we observed a statistically significant positive correlation between perceived explanation clarity and trust in the system ($\rho = 0.493, p = 0.005$). This suggests that the interpretability of model outputs remains a central factor in user acceptance of AI-assisted decisions, even when underlying model accuracy is high (accuracy = 95.8%, F1 = 0.96).

In this way, RG 1 is addressed not only at the model-performance level but also in a trust-sensitive evaluation setting.

7.1 Explanation Method Performance

Participants' strong preference for the perturbation-based method (LIME, Method 2) highlights the value of intuitive, word-level visualizations for threat moderation. This aligns with Poursabzi-Sangdeh et al. [47], who found that simpler, localized explanations can improve users' reasoning about model outputs. These results speak directly to RG 2 by providing a within-subject, realistic comparison of multiple explanation types under identical task conditions.

Latency concerns raised by several participants indicate that computational cost remains a barrier to real-time deployment, contributing to RG 3 by emphasizing the need for responsiveness in trustworthy moderation interfaces [38].

The limited utility of attention-based visualizations (Method 3) goes beyond user unfamiliarity and reflects structural weaknesses in raw attention interpretability. Prior critiques [57, 70] note that attention weights may not align with causal reasoning; in our study, this manifested as misleading salience on low-value tokens. Including Method 3 served as both a baseline and a targeted test of these critiques in a high-stakes setting, further advancing RG 2 by examining diverse explanation strategies—even those with known limitations—in realistic conditions. In practice, pairing attention with plain-language legends and token filtering aligns with broader principles of *usable transparency*, ensuring that explanation artifacts are understandable and actionable rather than merely available [65].

7.2 Trust, Usability, and Moderation Confidence

Trust and perceived moderation effectiveness were highly correlated ($\rho = 0.62, p < 0.01$), supporting Jacovi and Goldberg's [33] assertion that trust is closely tied to perceived task success. This addresses RG 2 by linking explanation clarity and user trust to measurable, task-level outcomes in a high-stakes, web-based moderation setting.

Usability emerged as a necessary but not sufficient condition for trust. The TFI received high usability ratings (mean = 4.27), yet trust varied more strongly with explanation clarity than with ease of use. This distinction reinforces that usability and interpretability are orthogonal design dimensions [41], directly informing RG 3 by suggesting that interface design must explicitly support sense-making, not only efficient interaction.

7.3 Human Factors in Moderation

7.3.1 Cognitive Effort and Task Complexity. Interaction-time analysis showed that message difficulty significantly influenced cognitive effort, except in the “easy” judicial-threat category, where participants still spent more time despite lower classification complexity. This likely reflects verification behavior: when content appears clearly severe, moderators may extend review time to confirm correctness—consistent with confirmation-bias effects in decision-making [46]. In high-stakes contexts, even clear-cut cases can prompt additional scrutiny due to perceived risk.

This finding supports RG 2 by illustrating how real-world ambiguity and perceived severity shape explanation use, even when

objective task difficulty is low. From a design standpoint, it contributes to RG 3 by suggesting tiered interaction modes: latency-sensitive “quick review” flows for routine cases and richer “deep review” flows for high-impact or ambiguous content, potentially combined with explicit system-confidence cues to help moderators balance speed and thoroughness.

7.3.2 Demographic Effects. Education level correlated positively with usability satisfaction: 83% of PhD respondents rated the system as “Excellent” or “Best Imaginable,” compared to 50% of Bachelor's degree holders ($\chi^2(6, N = 31) = 20.99, p < 0.05$). No significant differences appeared in trust or explainability ratings by gender, age, or prior AI/ML tool experience. This contrasts with domains where AI familiarity improves trust calibration [19, 29], suggesting that moderation performance may depend more on domain-specific knowledge (e.g., platform policies, threat typologies) than general AI literacy. Several non-technical participants with prior moderation experience reported high confidence interpreting explanations, while technically skilled but moderation-inexperienced participants described occasional uncertainty.

This outcome further supports RG 2 by indicating that domain-specific familiarity—not just AI literacy—shapes moderator performance and satisfaction. It also has practical implications for RG 3, underscoring the value of onboarding and training that emphasize platform policies and threat taxonomies, rather than relying solely on generic AI education.

7.4 Design Implications

Our results yield actionable guidelines for explainable moderation tools, representing direct outputs of RG 3 as concrete, deployable recommendations:

- **Prioritize simple, local visualizations.** Participants responded best to visualizations that clearly highlight relevant words and avoid visual clutter. Designs that make importance cues immediately visible scored highest for both trust and usability.
- **Treat latency as a first-class design constraint.** Even strong explanations lose value if slow to load. The slower response time of the preferred method (LIME) suggests that in fast-paced moderation, latency can directly affect engagement, workflow continuity, and ultimately trust.
- **Adopt hybrid explanation strategies.** Some participants proposed combining the clarity of word-level highlights with lightweight visual overviews. Hybrid approaches may mitigate the limitations of individual methods and better serve heterogeneous user needs.
- **Match explanation depth to risk.** For legally sensitive or high-impact content, moderators often spend extra time verifying even straightforward cases. Providing tiered modes—such as quick review for routine items and deep review for critical or ambiguous ones—can help align explanation depth and interaction time with task criticality.

By operationalizing explanation clarity, trust, and moderator usability in a live web context, TRuST-M advances *algorithmic transparency and fairness* [22] objectives for systems deployed on social-network platforms, in line with broader efforts toward interpretable, human-aligned AI on the social web.

7.5 Limitations and Future Work

This study involved a modest participant pool ($n = 31$), with only two professional moderators, which limits generalization to expert-heavy or large-scale operational environments. The underrepresentation of domain experts affects RG 2 and RG 3, and broader evaluation is needed to validate these findings. In future work, we plan to scale to cohorts of > 50 moderators across multiple platforms to strengthen external validity and expert coverage.

The fixed-order presentation of six messages, while providing consistent difficulty progression, may have introduced order effects. Future studies should use randomized or counterbalanced designs and randomize both message order and the mapping of anonymized XAI labels to methods to remove residual framing effects.

Demographic analysis showed that higher education levels were associated with greater usability satisfaction, while prior AI or moderation-tool experience did not significantly affect trust or explainability ratings. This diverges from other domains [19, 29], and targeted recruitment of professional moderators could clarify whether cognitive framing, risk perception, or task-specific heuristics drive this pattern.

Our saliency, perturbation, and attention-based explanations are complementary to other interpretability families, such as logic-based or symbolic surrogates that translate black-box reasoning into human-understandable rule sets [25]. Integrating such neuro-symbolic or rule-inductive approaches in moderation UIs may enhance global interpretability and policy alignment, but this remains outside the current experimental scope.

Several participants noted slower response times for the perturbation based method (LIME, Method 2), highlighting the need to explicitly assess latency in real-time workflows. Attention-based visualizations (Method 3) consistently underperformed in perceived usefulness and clarity, likely due to misleading emphasis on non-semantic tokens—a limitation documented in prior NLP interpretability research [34, 57, 70]. Its inclusion here served as both a baseline and an empirical validation of these critiques in high-stakes moderation (RG 2). Future work should quantify latency impacts, optimize computation pipelines, and refine attention-based methods via contextual weighting, token filtering, or hybrid strategies.

Finally, as this work was conducted in a controlled, single-session lab setting, longitudinal field deployments in operational environments are needed to assess how trust, accuracy, and explanation preferences evolve with sustained, high-volume, and contextually diverse threat exposure. Such deployments would extend RG 3 by examining how design guidelines hold up under real-world workload, policy change, and adversarial adaptation.

8 Conclusion

This paper introduced TRuST-M, a framework for evaluating how explanation methods shape user trust, usability, and perceived effectiveness in LLM-based threat moderation. Using a domain-adapted RoBERTa classifier and three complementary explanation techniques—gradient-based saliency, LIME, and attention visualization—we conducted a within-subjects study to examine moderation decisions in a social-web setting. Explanation clarity showed a significant positive correlation with trust ($\rho = 0.493$, $p = 0.005$), LIME’s localized highlights were preferred by 58% of participants

despite latency costs, and trust aligned closely with perceived moderation effectiveness ($\rho = 0.62$, $p < 0.01$). Usability alone did not guarantee trust, underscoring the importance of interpretable, cognitively lightweight visualizations.

We recommend prioritizing explanation formats that minimize cognitive load, reducing latency for real-time workflows, and exploring hybrid strategies that combine complementary strengths. Future work should validate these findings with larger and more expert-diverse cohorts, and through longitudinal deployments in operational moderation pipelines to refine explanation design and establish standards for transparent, accountable AI. Extending TRuST-M toward fairness-aware and globally interpretable evaluation represents a promising direction for aligning automated moderation tools with real-world policy and practitioner needs.

Overall, TRuST-M positions explainability as an *interpretable decision-support layer* for web and social-network moderation, aligned with ongoing efforts to build transparent, trustworthy, and responsibly deployed AI systems.

Ethical Considerations

This study received University IRB approval (ID: STUDY0000ZZ00) under secondary data analysis. All model-development data were drawn from publicly accessible Telegram channels without collecting personally identifiable information. The dataset contained both far-right and far-left sources, though far-right content was more prevalent; this imbalance is acknowledged as a limitation (see *Data* section).

Participants were briefed using a neutral demonstration message, provided an explicit content warning, and were encouraged to ask questions throughout to safeguard well-being. Interaction logs, timing data, and survey responses were anonymized and stored securely on university systems. All anonymized data and analysis code will be released to support reproducibility (see *Procedure* section).

While TRuST-M aims to strengthen trust and usability in AI-assisted moderation, we note potential risks, including over-reliance on automated outputs or inadvertent suppression of lawful speech and emphasize the need for human oversight in deployment (see *Moderation Effectiveness*). For real-world implementations, we recommend integrating lightweight bias-visualization modules [14] and adopting reproducible evaluation and unlearning checklists to ensure ongoing fairness and accountability [15, 21].

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